ROYAL SOCIETY OPEN SCIENCE

accepted

R. Soc. open sci.
doi:10.1098/not yet assigned

Supplementary Information for

A Retrospective Assessment of COVID-19 Model Performance in the US

Kyle J. Colonna[a]*, Gabriela F. Nane[b], Ernani F. Choma[a], Roger M. Cooke[b,c], and John S. Evans[a]

[a] *Environmental Health Department, Harvard T.H. Chan School of Public Health, Harvard University, Boston, MA 02115, USA*
[b] *Department of Mathematics, Delft University of Technology, Delft, 2628 XE, The Netherlands*
[c] *Resources for the Future, Washington, DC 20036, USA*

Keywords: COVID-19; Decision Sciences; Forecasting; Uncertainty Analysis; Cooke's Classical Model; Ensemble Modelling

1. Technical Details of the Classical Model

The classical model for evaluating and combining experts considers experts as statistical hypotheses and prioritizes statistical accuracy. Informativeness is also important and serves to modulate between experts with similar statistical accuracy so that it is impossible to compensate poor statistical accuracy with very high information. Performance weights for combining experts are derived which satisfy asymptotic strictly proper scoring rule constraints. Quantile averaging is briefly discussed, but Bayesian approaches are omitted [1], as they have not been applied to model based COVID forecasting

The current study considered five assessed quantiles (i.e., *5%*, *25%*, *50%*, *75%* and *95%*) for each elicited uncertain quantity, or item. The expert could be a human assessor, a computer code, or some combination of the two.

1.1. Statistical Accuracy (aka Calibration)

The assessed quantiles divide the range of possible values into six inter-quantile intervals for which an expert's probabilities are known (e.g., $p_1 = 0.05$ [less than or equal to the *5%* quantile]; $p_2 = 0.20$ [greater than the *5%* quantile and less than or equal to the *25%* quantile], etc.). If N quantities are assessed as calibration questions, or items, each expert may be regarded as a statistical hypothesis, namely that each realization falls in one of the six inter-quantile intervals with the probability vector:

$$p = (0.05, 0.20, 0.25, 0.25, 0.20, 0.05)$$

Suppose we have realizations $x_1, \ldots, x_N$ of these quantities. We may then form the sample distribution of the expert's inter-quantile intervals as $s_e = (s_{1,e}, s_{2,e}, \ldots, s_{6,e})$, where:

$$s_{1,e} = \#\{i \mid x_i \leq 5\% \text{ quantile}\} / N$$

$$s_{2,e} = \#\{i \mid 5\% \text{ quantile} < x_i \leq 20\% \text{ quantile}\} / N$$

$$\ldots$$

$$s_{6,e} = \#\{i \mid 95\% \text{ quantile} < x_i\} / N$$

*Author for correspondence (kcolonna@g.harvard.edu)

Note that the sample distribution depends on the assessments from the expert, *e*. If the realizations are indeed drawn independently from a distribution with quantiles as stated by the expert, then the quantity:

$$2N \sum_{i=1}^{6} s_{i,e} \ln\left(\frac{s_{i,e}}{p_i}\right) \tag{1.1}$$

is asymptotically distributed as a chi-square random variable with five degrees of freedom. This is the likelihood ratio statistic. If we extract the leading term of the logarithm, we obtain the familiar chi-square test statistic for goodness of fit. If, after a few realizations, the expert saw that all realizations fell outside of their *90%* central confidence interval, (s)he might conclude that their intervals were too narrow and broaden them on subsequent assessments. This means that, for this expert, the uncertainty distributions are *not* independent, and they learn from the realizations. Expert learning is not a goal of a study, and their joint distribution is not elicited. Rather, the analyst wants experts who do not need to learn from the elicitation. Hence the analyst scores the expert, *e*, as the statistical likelihood of the hypothesis:

H_e: The inter-quantile interval containing the true value for each item is drawn independently from probability vector, p.

A simple test for this hypothesis uses the test statistic (1.1), and the likelihood, or p-value, or **statistical accuracy score** (aka **'calibration score'**) of this hypothesis, is:

$$C_e = P\,(\chi \geq r \mid H_e) \tag{1.2}$$

where χ is a chi-square distributed random variable with five degrees of freedom and *r* is the value of (1.1) based on the observed values $x_1, \ldots, x_N$. It is the probability under hypothesis H_e that a deviation at least as great as *r* should be observed on *N* realizations if H_e were true. Calibration scores are absolute and can be compared across studies. However, before doing so, it is appropriate to equalize the power of the different hypothesis tests by equalizing the effective number of realizations. To compare scores on two data sets with *N* and *N'* realizations, we simply use the minimum of *N* and *N*' in (1.1), without changing the sample distribution, *s*.

Although the calibration score employs the language of simple hypothesis testing, it must be emphasized that we are not rejecting expert-hypotheses; rather we are using this language to measure the degree to which the data supports the hypothesis that the expert's probabilities are statistically accurate. Low scores, near zero, mean that it is unlikely that the expert's probabilities are correct. A common thumb rule in testing a simple multinomial hypothesis is that there should be *5* expected observations in each cell. In practice this is often relaxed to one expected observation. With *10* calibration variables, there is *½* expected observation in the lowest and highest cells. Simulation studies [2] show that the statistical power afforded by *10* observations enables only the identification of gross differences in statistical performance. The default choice of *10* calibration variables reflects a compromise allowing coarse distinctions without excessively burdening the elicitation. This compromise is motivated by the fact that large differences are usually present in expert panels [3], bearing in mind that CM is not testing statistical hypotheses as such.

1.2. Information

Measuring information requires associating a density to each assessment for each item. To do this, we use the unique density that complies with the expert's quantiles and is minimally informative with respect to a background measure. For a uniform background measure, the density is constant between the assessed quantiles, and is such that the total mass between the quantiles agrees with *p*. The background measure is not elicited from experts as it must be the same for all experts; instead, it is chosen by the analyst. The Classical Model (CM) uses by default either the uniform or the log-uniform background measure.

The uniform and log-uniform background measures require an 'intrinsic range' on which these measures are concentrated. The CM implements the so-called '*k% overshoot rule*': for each item, we consider the smallest interval, $I = [L, U]$, containing all the assessed quantiles of all experts and the realization, if known. This interval is extended to:

$$I^* = [L^*, U^*]; L^* = L - k\,(U - L) / 100; U^* = U + k\,(U - L) / 100 \tag{1.3}$$

The value of k is chosen by the analyst. A large value of k tends to make all experts look quite informative and tends to suppress the relative differences in information scores. The **information score** of the expert, e, on assessments for uncertain quantities $1, \ldots, N$, is:

$$Inf_e = (1 / N) \sum_{i = 1, \ldots, N} I\,(f_e, i \mid g_i) \tag{1.4}$$

where g_i is the background density for item i and $f_{e,i}$ is the fitted density for expert, e, for item i. This is proportional to the relative information of the expert's joint distribution given the background, under the assumption that the items are independent. As with calibration, the assumption of independence reflects a desideratum of the analyst and is not an elicited feature of the expert's joint distribution. The information score does not depend on the realizations. An expert can acquire a high information score by choosing their quantiles very close together.

Evidently, the information score of an expert depends on the intrinsic range and on the assessments of the other experts. Hence, information scores cannot be rigorously compared across studies.

Both statistical accuracy and informativeness are dimensionless. The information score is a 'slow' function, that is, large changes in the assessments produce only modest changes in the information score. Information scores in a panel of experts typically vary by a factor less than *3* whereas the statistical accuracy (1.1) varies over several orders of magnitude. This ensures that the normalized product of statistical accuracy and information is driven by the calibration score. It also means that modest changes in informativeness correspond to sizeable changes in the distributions. Increasing informativeness by a factor of two roughly corresponds to halving the distance between the 95th and 5th percentiles.

1.3. Scoring Rules for Individual Items

Scoring rules were introduced by de Finetti in 1937 as tools for encouraging honesty in eliciting subjective probabilities [4] and have been further developed by many authors [5]. An expert receives a score as a function of their probability assessment and the realization. The score is strictly proper if the expert maximizes the expected score per item by, and only by, stating their true belief. Using a result of Murphy (1977) [6], DeGroot and Fienberg (1983) gave an additive decomposition of strictly proper rules into 'calibration' and 'refinement' terms [7], thereby replacing Murphy's 'resolution' (refinement applies only to well-calibrated experts). In the case of the logarithmic rule, refinement becomes Kullback-Leibler Divergence (aka, directed divergence, cross entropy, or relative information [the term we use]).

Scoring rules for individual variables were not designed for evaluating or combining experts and are not generally fit for that purpose. Indeed, rewarding honesty is not the same as rewarding quality. A simple example explains this difference: Consider *100*-coin tosses. An expert assesses the probability of heads on each toss as ½. The score for the outcome heads is the same as their score for tails on each toss. If the score for all *100* assessments is a function of their *100* scores for the individual tosses, then their score for *100* tosses is independent of the outcome sequence; the outcome of *100* heads receives the same score as *50* heads and *50* tails.

Another example concerns the quadratic rule for '*rain / no rain*'. This rule is positively sensed on *[-1.1]* and assigns the score $2r - r^2 - (1 - r)^2$ if rain occurs and r is the expert's probability of rain. Interchange r and $(1 - r)$ in case it doesn't rain. Consider *1000* next day forecasts of rain by two experts. The experts bin their forecasts as shown below [2]:

Probability of Rain next day:		5%	15%	25%	35%	45%	55%	65%	75%	85%	95%	Totals
expert 1	assessed	100	100	100	100	100	100	100	100	100	100	1000
	realized	5	15	25	35	45	55	65	75	85	95	500
expert 2	assessed	100	100	100	100	100	100	100	100	100	100	1000
	realized	0	0	0	0	0	100	100	100	100	100	500
average quadratic score	expert 1	0.67										
	expert 2	0.84										

The experts are equally informative in the sense that they assign the same probabilities to the same number of days. *Expert 1* is statistically perfectly accurate whereas *expert 2* is massively inaccurate. Woe unto anyone basing their decisions on *expert 2's* forecasts. Nonetheless, *expert 2's* quadratic score is higher than that of *expert 1*. For more discussion see [2,8,9].

The Probability Interval Score *(PIS)* and its related Continuous Ranked Probability Scores (*CRPS)* have recently been applied to COVID-19 probabilistic predictions [10,11] so it is appropriate to include a brief discussion of these. Numerical insight into these scores requires a bit more effort, so more detail is provided.

For the $(1-\alpha)$ interval $[L, U]$ with upper (lower) bound U (L), the *PIS* (negatively sensed) for realization y is $(U - L) + (2 / \alpha) \times [(L - y)_+ + (y - U)_+]$ where $X_+ = X$ if $X > 0$ and $= 0$ otherwise. $s = 2 / \alpha$ is the slope of the overconfidence penalty for $Y \notin [L, U]$. The length $(U - L)$ is called the 'sharpness'; small values reward concentrated probability mass. If $Y \sim Unif[0, 1]$, the central 0.9 interval is $[0.05, 0.95]$ with expected *PIS*:

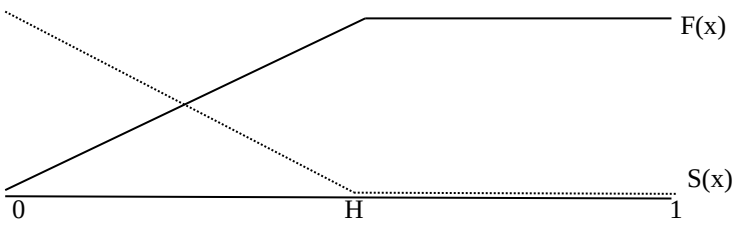
$$0.9 + 2 \times \int_{0 \ldots 0.05} s \times u \, du = 0.9 + (2 / 0.1) \times 0.05^2 = 0.95$$

The integral is doubled to account for $Y > U$. Suppose an expert prefers to give an *80%* interval $[0.1, 0.9]$, $s = 2 / 0.2 = 10$. The expected score is $0.8 + 2 \times \int_{0 \ldots 0.1} s \times u \, du = 20 \times ½ \times 0.1^2 = 0.9 < 0.95$. An expert seeking to optimize (i.e., minimize) their expected score might take a central *2%* prediction interval $[0.49, 0.51]$ with expected score $0.02 + 2 \times 2 / 0.98 \times 0.49^2 / 2 = 0.51$ (or take $\lim_{\epsilon \to 0} [,5 - \epsilon, 5 + \epsilon]$ with expected score ½). All these prediction intervals have zero information relative to the uniform background measure on $[0, 1]$, so from that viewpoint there isn't much to choose.

The way in which the *PIS* trades overconfidence for sharpness may strike some as counter intuitive. For example, an expert claiming the degenerate interval $[0.5, 0.5]$ has *40%* probability of catching the realization would achieve an expected score of *0.833*, better than the score of the central *90%* interval. The sharpness of an interval of zero length outweighs the overconfidence of claiming *40%* mass at the point *0.5*. Of course, this example is blocked if probability intervals are required to be *90%*; assigning *90%* mass to the point *0.5* returns an interval score of *5*. Such scores from several experts could cause bad statistical performance, depending on how the experts are combined.

The weighted *PIS* converges to the continuous ranked probability score (*CRPS*). Applied to a set of probabilistic forecasts and realizations, the expected *CRPS* is based not on what the expert believes, but on the realizations. For illustration we assume that realizations are sampled from variable Y uniformly distributed on $[0, 1]$.

Consider an assessment of Y by an expert whose distribution is $X \sim Unif[0, H]$, $H \leq 1$. The expert thinks values above H are impossible, although these can in fact arise. The *CDF* of X, $F(x) = (x / H)$ and the survivor function of X, $S(x) = (1 - CDF$; dotted) are pictured below:



The expected *CRPS* is: $\int_{y=0\ldots1}\int_{x=0\ldots1}(F(x)-1_{\{x\geq y\}})^2\,dx\,dy$. The calculation is broken into 2 steps:

A) $y \leq H$:

$$\int_{y=0\ldots H}\left[\int_{x=0\ldots y}(x/H)^2\,dx+\int_{x=y\ldots H}((H-x)/H)^2\,dx\right]dy$$

$$=\int_{y=0\ldots H}\left[y^3/(3H^2)+\int_{z=H-y\ldots0}z^2/H^2\,(-dz)\right]dy=$$

$$=\int_{y=0\ldots H}\left[y^3/(3H^2)+(H-y)^3/(3H^2)\right]dy$$

$$=H^2/12+(1/(3H^2))\int_{z=H\ldots0}z^3\,(-dz)=H^2/6$$

B) $y>H$:

$$\int_{y=H\ldots1}\left[\int_{x=0\ldots H}(x/H)^2+\int_{x=H\ldots y}dx+\int_{x=y\ldots1}0\,dx\right]dy$$

$$=\int_{y=H\ldots1}[H/3+y-H]\,dy$$

$$=H(1-H)/3+\int_{0\ldots1-H}z\,dz$$

$$=H(1-H)/3+(1-H)^2/2$$

Therefore:

$$E(CRPS(F,Y))=H^2/6+H(1-H)/3+(1-H)^2/2$$

Notice that the result for (A) is the score an expert with $X\sim Unif[0,H]$ expects, namely $H^2/6$, which has the physical dimension of H^2. If X is in meters and changes to centimeters, the expected score increases by a factor 10^4. Note also that if $\{F_i\}\rightarrow F$ in distribution then $E(CRPS(F_i,Y)\rightarrow E(CRPS(F,Y)$ by the Helly Bray theorem.

If $X\sim Unif[L,H]$, $0\leq L\leq H\leq 1$, then the same method of calculation applies mutatis mutandis. If $L=1-H$, $H\geq 0.5$, then the contributions from $x\leq y$ and $y\leq x$ are equal, and we need only double the contribution from $x\leq y$. In that case, for $y\leq L$, the contribution from $x\leq y$ is zero, since $x>L$. We compute:

$$\int_{y=L..H}\int_{x=L\ldots y}F(x)^2dx\,dy+\int_{y=H..1}\int_{x=L\ldots y}F(x)^2dx\,dy$$

$$=\int_{y=L..H}\int_{x=L\ldots y}(x-L)^2/(H-L)^2dx\,dy+\int_{y=H..1}\left[\int_{x=L\ldots H}(x-L)^2/(H-L)^2dx+\int_{x=H\ldots y}dx\right]dy$$

$$=\int_{y=L..H}(y^3/3)/(H-L)^2dx+\int_{y=H\ldots1}[(H-L)/3+(y-H)]\,dy$$

$$=(H-L)^2/12+(H-L)(1-H)/3+(1-H)^2/2$$

Adding the identical contribution from $y\leq x$ gives:

$$E(CRPS(F,y))=(H-L)^2/6+2(H-L)(1-H)/3+(1-H)^2$$

Again, the score inherits a physical dimension from X. Substituting $L=1-H$ we find that the expected score for $X\sim U[0,H]$, is equal that of $X\sim U[1-H,H]$: For $X\sim Unif[0,0.7]$, $E(CRPS(F,y))=0.196\underline{6}=E(CRPS(F',y))$ for $X'\sim Unif[0.3,0.7]$. By the same token, an expert who believes $X\sim[0,0.5]$ gets the same expected score as one who concentrates all mass in the point 0.5, namely $\frac{1}{4}$. By the Helly Bray theorem, if F_i converges in distribution $Unif[0,0.5]$ and F'_i converges to the distribution with mass concentrated at 0.5, then for all $\varepsilon>0$ and all sufficiently large n, $|E(CRPS(F_n,Y))-E(CPRS(F'_n,Y))|<\varepsilon$. Again, this illustrates how the *CRPS* compensates loss of statistical accuracy with a gain in "sharpness" and again presents challenges in explaining the scores to experts and decision makers.

1.4. Scoring Rules for Average Probabilities

To avoid problems with scores for individual items, Cooke introduced scoring rules for average probabilities [8]. Let random variables $X_1, ..., X_n$ take outcomes in a finite set, O, let M_O be the set of probability measures on O, and let M_n be the set of probability measures on $X_1, ..., X_n$. For $\Pi \in M_n$, let π be the vector of average probabilities, that is, $\pi_i = (1 / n) \Sigma_{j = 1, ..., n} \Pi(X_j = i)$. Let s be the observed relative frequency of outcomes for realization, $(X_1, ..., X_n) = (x_1, ..., x_n)$. A scoring rule for average probabilities assigns a number, R, to the pair (π, s). R is strictly proper (positively sensed) if:

$$\text{for all } \Pi \in M_n, \text{argmax}_{\varphi \in Mo} E_{\Pi}(R(\varphi, s)) = \pi$$

This says, whatever the expert's belief, Π, about $(X_1, ..., X_n)$, they achieve the maximal expected score by stating the probability, π, over outcomes which corresponds to their average probabilities. The proofs are a bit more complicated because, "*for all* Π," goes over a much larger set than the *argmax* over M_o.

There is a representation theorem in Cooke (1991) for such rules [8]. However, more useful in practice are rules which are asymptotically strictly proper as $n \to \infty$. These rules allow the product form in the CM.

Familiarity with foundations teaches that the problem of combining experts' judgments is not a mathematical problem. The laws of probability even supplemented with Savage's axioms and the theory of proper scoring rules, will never tell us how to combine experts. The problem is more akin to finding an optimal design in engineering. For example, a bicycle obeys Newton's laws but doesn't follow from them. Any working design will involve features motivated by practicalities rather than laws. So let it be with measures of 'spread'. Traditional measures like the standard deviation and prediction intervals are unsuitable because they inherit the physical dimension of the underlying variables: changing from meters to kilometres changes the numbers. To compare spreads across variables with different physical dimensions we need a measure which is scale invariant. We also need it to be 'tail insensitive' because the tails in expert judgment are poorly constrained. Finally, it must be 'slow'.

The theory of asymptotic proper scoring rules for average probabilities gives a product form of *measure of statistical accuracy* $\times$ *measure of lack of spread*. Weights will be formed by normalizing such products. We want statistical accuracy (aka, calibration; a very fast function) to dominate, therefore the measure of "*lack of spread*" must be slow. Relative information fits the bill and requires specifying 'relative to what'. This is as it should be because information in a distribution is always relative to another (background) distribution.[i]

Since the background distribution must integrate to one, it must be 'concentrated' somewhere. The uniform or log-uniform distributions are chosen as default backgrounds because they have no location parameter other than the endpoints of their support. However, we do need to specify the compact support of the background for each item. We do this by choosing the smallest interval containing all assessments and the realization, if available, plus a *k%* overshoot. If k is very large then all experts appear very informative relative to the background, and this tends to discount the role of information in determining weights. The CM dictates that the choice of uniform/log-uniform and k are choices of the analyst, and any such choices must be controlled by user parameters in the code. Thus, we can see that k must be very large to make much difference. Uniform vs log-uniform can have an influence; the guidance is that if we reason about orders of magnitude – use log-uniform.

2. Combining Experts

We discuss quantile averaging and linear pooling based on performance weights.

[i] You can see this in the continuous from of the entropy integral. For a discrete distribution $P = p1, ..., pn$, the entropy is defined as $H(P) = - \Sigma pi\ ln(pi)$. To pass to the continuous version replace $\Sigma \to \int$ and $pi \to f(x)\ dx$. That gives $H(f) = -\int fx\ (dx)\ ln(f(x)\ dx)$, which is meaningless. You must use relative information $I\ (f \mid g) = \int f(x)\ dx\ ln(f(x)\ dx / g(x)\ dx)$ so that the dx's inside the ln cancel and $I(f \mid g) = \int f(x)\ dx\ ln(f(x) / g(x))$.

2.1. Quantile averaging

Many practitioners combine expert judgments by quantile averaging. This is mostly done unawares, but the method has also been promoted as an alternative to equal weighting of distributions [12]. It has been studied in [9, 13] and compared with expert judgment data [14]. This publication establishes a variance shrinkage law:

$$Var\ Equal\ Weight\ combination = Var\ of\ means + Ave\ Var \geq Ave\ Var \geq Var\ Quantile\ Average$$

In as much as the equal weight combination is generally statistically accurate, this result implies that quantile averaging risks overconfidence, especially if there is a large difference in experts'/models' means. A recent tally of 58 post-2006 expert judgment studies showed that in 28 studies the hypothesis that the quantile average was statistically accurate would be rejected at the *5%* level, and in 15 studies rejection was at the 0.1% level [9].

2.2. Linear pooling with performance weights in the Classical Model

The 'combined score', *Cs*, of the expert, *e*, is the dimensionless quantity*:*

$$Cs_e = C_e * Inf_e \tag{2.1}$$

A scoring rule is (asymptotically) strictly proper if an expert achieves their (long run) maximal expected score by, and only by, giving assessments corresponding to their true beliefs. That is, an expert maximizes their long run expected score by, and only by, ensuring that the probabilities, $p = (0.05, 0.20, 0.25, 0.25\ 0.20, 0.05)$, correspond to their true beliefs. The theory of proper scoring rules tells us that (2.1) becomes a proper scoring rule if it is augmented with a cut-off, α, on the calibration score such that an expert is unweighted if $C < \alpha$. The α is like the significance level in simple hypothesis testing, but its purpose is different. The goal is to measure 'goodness' with a strictly proper scoring rule.

A combination of expert assessments is called a '*decision-maker*' (*DM*). All *DMs* discussed here are examples of linear pooling. The Classical Model is essentially a method for deriving weights in a linear pool. A 'good probabilistic expert' corresponds to an expert with good calibration (i.e., high statistical likelihood; high p-value) and high information. We want weights which reward good experts, and which pass these virtues on to the *DM*.

The reward aspect of weights is very important. We could simply solve the following optimization problem: find a set of weights such that the linear pool under these weights maximizes the combined score of the *DM*. When solving this problem with real data, one finds that the weights do not generally reflect the performance of the individual experts. As we do not want an expert's influence on the *DM* to appear haphazard, and we do not want to encourage experts to game the system by tilting their assessments to achieve a desired outcome; we must impose a strict scoring rule constraint on the weighting scheme.

The scoring rule constraint requires that the combined score is multiplied by the indicator function, $1_\alpha(C_e \geq \alpha)$, which takes the value *1* if $C_e \geq \alpha$ and *0* otherwise*:*

$$w_{\alpha,e} = C_e * Inf_e * 1_\alpha(C_e \geq \alpha) \tag{2.2}$$

This says that the expert, *e*, is weighted only if their calibration is at least α. The resulting *DM* is a function of α:

$$DM_{\alpha,\,i} = \sum_{e\,=\,1,\,\ldots,\,E} w_{\alpha,\,e}\, f_{e,\,i} \,/\, \sum_{e\,=\,1,\,\ldots,\,E} w_{\alpha,\,e} \tag{2.3}$$

Scoring rule theory does not say what the value of α should be. In practice there are three ways for choosing α. The '*optimized*' *DM* chooses α such that the resulting combined score of the *DM* is maximized. The *optimized DM* is DM_{α^*} where α^* maximizes $C(DM_\alpha) \times Inf(DM_\alpha)$. This typically leads to choosing an α so high that only one or two

experts are weighted. The '*statistical threshold*' *DM* chooses an α to distribute weight over experts with 'acceptable' calibration (typically, $\alpha = 0.05$ or 0.01). The '*inclusive*' *DM* chooses an α so low that all experts are weighted. If this choice is made a posteriori, then this is not a strictly proper scoring rule. It is to be noted that unweighted experts still have influence over the intrinsic range, and their rationales are recorded.

These weights are termed global because the information score is based on all the assessed calibration items. A variation on this scheme allows a different set of weights to be used for each item. This is accomplished by using information scores for each item rather than the average information score:

$$w_{\alpha,\, e,\, i} = 1_{\alpha}(C_e \geq \alpha) * C_e * f_{e,\, i}\, ln\, (f_{e,\, i} \mid g_i) \tag{2.4}$$

For each α we define the '*item weight*' DM_{α} for the item, i, as:

$$IDM_{\alpha,\, i} = \sum_{e\, =\, 1,\, ...,\, E} w_{\alpha,\, e,\, i}\, f_{e,\, i} \,/\, \sum_{e\, =\, 1,\, ...,\, E} w_{\alpha,\, e,\, i} \tag{2.5}$$

The same variation applies to the *threshold DM* and the *inclusive DM*.

Item weights are potentially more attractive as they allow experts to up- or down-weight themselves for individual items according to how much they feel they know about that item. 'Knowing less' means choosing quantiles further apart and lowering the information score for that item. Of course, the 'good performance' of item weights requires that experts can perform this up/down-weighting successfully. Anecdotal evidence suggests that item weights improve over global weights as the experts receive more training in probabilistic assessment. For both global and item weights, calibration dominates over information; information serves to modulate between equally well calibrated experts. Definitions and proofs of these scoring rule properties are found in Cooke (1991) [1].

Optimizing the weights in (2.2) and (2.4) often causes experts to be unweighted, even those with good scores. Such experts are not 'rejected'; unweighting simply means that their input is already captured by a smaller subset of experts. Their value to the whole study is brought out in studying the robustness of the *optimal DM* under the loss of experts and in determining the intrinsic range.

Supplemental References

1. Rougier J, Goldstein M, House L. 2013 Second-Order Exchangeability Analysis for Multimodel Ensembles. *Journal of the American Statistical Association* **108**, 852–863. (doi:10.1080/01621459.2013.802963)
2. Cooke RM. 2014 Validating Expert Judgment with the Classical Model. In *Experts and Consensus in Social Science* (eds C Martini, M Boumans), pp. 191–212. Cham: Springer International Publishing. (doi:10.1007/978-3-319-08551-7_10)
3. Cooke RM, Marti D, Mazzuchi T. 2021 Expert forecasting with and without uncertainty quantification and weighting: What do the data say? *International Journal of Forecasting* **37**, 378–387. (doi:10.1016/j.ijforecast.2020.06.007)
4. de Finetti B. 1937 La prévision : ses lois logiques, ses sources subjectives. *Annales de l'institut Henri Poincaré* **7**, 1–68.
5. Shuford EH, Albert A, Edward Massengill H. 1966 Admissible probability measurement procedures. *Psychometrika* **31**, 125–145. (doi:10.1007/BF02289503)
6. Murphy AH. 1977 The value of climatological, categorical, and probabilistic forecasts in the cost-loss ratio situation. *Monthly Weather Review* **105**, 803–816.
7. DeGroot MH, Fienberg SE. 1983 The comparison and evaluation of forecasters. *The Statistician* **32**, 14–22.
8. Cooke RM. 1991 *Experts in Uncertainty: Opinion and Subjective Probability in Science*. Oxford, UK: Oxford University Press.
9. Colson AR, Cooke RM. 2017 Cross validation for the classical model of structured expert judgment. *Reliability Engineering & System Safety* **163**, 109–120. (doi:10.1016/j.ress.2017.02.003)
10. Ray EL *et al.* 2020 Ensemble Forecasts of Coronavirus Disease 2019 (COVID-19) in the U.S. *medRxiv* , 2020.08.19.20177493. (doi:10.1101/2020.08.19.20177493)
11. Cramer EY *et al.* 2022 Evaluation of individual and ensemble probabilistic forecasts of COVID-19 mortality in the United States. *Proceedings of the National Academy of Sciences* **119**, e2113561119. (doi:10.1073/pnas.2113561119)
12. Lichtendahl KC, Grushka-Cockayne Y, Winkler RL. 2013 Is It Better to Average Probabilities or Quantiles? *Management Science* **59**, 1594–1611. (doi:10.1287/mnsc.1120.1667)
13. Bamber J, Aspinall W, Cooke R. 2016 A commentary on "how to interpret expert judgment assessments of twenty-first century sea-level rise" by Hylke de Vries and Roderik SW van de Wal. *Climatic Change* **137**, 321–328. (doi:10.1007/s10584-016-1672-7)
14. Cooke, Roger M. (2022) Averaging Quantiles, Variance Shrinkage and Overconfidence, Futures & Foresight Science published by John Wiley & Sons Ltd.https://onlinelibrary.wiley.com/doi/epdf/10.1002/ffo2.139

Supplemental Tables

Table S1. Dimensions, ProQuest Policy, and Factiva database search results

Model	Dimensions Database Results	ProQuest Policy Database Results	Manuscript Citations	Manuscript Altmetric	Manuscript Link	Factiva Database Results	Search Terms
BPagano-RtDriven	3	--	--	--	--	0	("Bob Pagano" or "BPagano") and (COVID or coronavirus)
CovidAnalytics-DELPHI	11	--	27	6	https://doi.org/10.1101/2020.06.23.20138693	23	("Massachusetts Institute of Technology" or "MIT" or "CovidAnalytics" or "Operations Research Center") and ("CovidAnalytics-DELPHI" or "Michael Lingzhi Li" or "Hamza Tazi Bouardi" or "Omar Skali Lami" or "Dimitris Bertsimas") and (COVID or coronavirus)
COVIDhub-baseline*	9	--				7	COVIDhub-baseline or ((COVIDhub or "COVID-19 Forecast Hub") and baseline)
CU-select*,**	9	11	64	33	https://doi.org/10.1101/2020.03.21.20040303	134	("Columbia University" or "CU" or "CUMC") and ("cu-select" or "Sen Pei" or "Shaman") and (COVID or coronavirus) and (forecast or predict or project)
DDS-NBDS	5	--	--	--	--	0	("Discrete Dynamical Systems" or "DDS-NBDS" or "Rahi Kalantari" or "Mingyuan Zhou") and (COVID or coronavirus)
epiforecasts-ensemble1	4	--	56	2	https://wellcomeopenresearch.org/articles/5-112/v2	27	("London School of Hygiene & Tropical Medicine" or LSHTM or epiforecasts) and (epiforecasts-ensemble1 or Sherratt or "Sam Abbott" or "Nikos Bosse" or "Sophie Meakin" or "Sebastian Funk") and (COVID or coronavirus)
GT-DeepCOVID	6	7	21	27	https://doi.org/10.1101/2020.09.28.20203109	1	("Georgia Tech" or "GT" or gatech) and (Aditya Prakash or Bijaya Adhikari or DeepCOVID) and (COVID or coronavirus)
IHME-CurveFit*	9	--	314	370	https://doi.org/10.1101/2020.03.27.20043752	approx. 4000	("Institute for Health Metrics and Evaluation" or "IHME" or "IHME-CurveFit") and (COVID or coronavirus) and (forecast or predict or project)
JHU_CSSE-DECOM	4	--	--	--	--	48	("Johns Hopkins" or "JHU" or "CSSE" or "Center for Systems Science and Engineering") and ("JHU_CSSE-DECOM" or "Maximilian Marshall" or "Lauren Gardner" or "Carson Drew" or "Erin Burman" or "Kristen Nixon") and (COVID or coronavirus)
JHUAPL-Bucky	8	--	--	--	--	1	("Johns Hopkins" or "JHU" or "JHUAPL") and ("JHUAPL-Bucky" or "Matt Kinsey" or "Tallaksen" or "Obrecht") and (COVID or coronavirus)
Karlen-pypm	12	--	0	2	https://arxiv.org/abs/arXiv:2007.07156	33	("Karlen-pypm" or "Dean Karlen") and (COVID or coronavirus)

MIT_CritData-GBCF	1	--	0	11	https://doi.org/10.1101/2021.11.06.21266007	7	("Massachusetts Institute of Technology" or "MIT" or "MIT Critical Data") and (MIT_CritData-GBCF or "Nicolás Della Penna" or "Nicolas Della Penna" or "Saketh Sundar" or "Leo Anthony Celi" or "Leo Celi") and (COVID or coronavirus)
MOBS-GLEAM_COVID	6	--	44	139	https://doi.org/10.1126/sciadv.abd6989	39	("Northeastern University" or "Laboratory for the Modeling of Biological and Socio-technical Systems" or "MOBS") and ("MOBS-GLEAM_COVID" or "Alessandro Vespignani" or "Matteo Chinazzi" or "Jessica T. Davis" or "Kunpeng Mu" or "Ana Pastore y Piontti") and (COVID or coronavirus) and (forecast or project or predict)
PSI-DRAFT	7	--	--	--	--	0	("Predictive Science" or "PSI") and ("PSI-DRAFT" or "James Turtle" or "Michal Ben-Nun" or "Pete Riley") and (COVID or coronavirus)
RobertWalraven-ESG	5	--	--	--	--	0	("Robert Walraven" or "RobertWalraven-ESG") and (COVID or coronavirus)
SteveMcConnell-CovidComple	4	--	--	--	--	0	("Steve McConnell" or "SteveMcConnell-CovidComplete" or "CovidComplete") and (COVID or coronavirus)
UCSD_NEU-DeepGLEAM	4	--	0	2	https://arxiv.org/abs/arXiv:2102.06684	0	("University of California" or "UCSD" or "Halıcıoğlu Data Science Institute" or "Halicioglu Data Science Institute" or "HDSI") and ("UCSD_NEU-DeepGLEAM" or "Dongxia Wu" or "Liyao Gao" or "Yian Ma" or "Rose Yu" or "Palaksh Vardhan Rungta" or "Xinyue Xiong" or "Matteo Chinazzi" or "Alessandro Vespignani") and (COVID or coronavirus)
UMass-MechBayes	8	--	5	1	https://doi.org/10.1101/2020.12.22.20248736	4	("University of Massachusetts" or "UMass" or "UMass-Amherst" or "University of Massachusetts-Amherst") and ("UMass-MechBayes" or "Dan Sheldon" or "Graham Gibson" or "Nick Reich") and (COVID or coronavirus)
USC-SI_kJalpha	4	--	--	2	https://arxiv.org/abs/2007.05180	2	("University of Southern California" or "USC") and ("USC-SI_kJalpha" or "Srivastava") and (COVID or coronavirus)

* Had multiple manuscripts; the manuscript with the most citations is detailed
** Four similar variations of this model were selected; the most relevant version is detailed

Table S2. Median predictive (i.e., accuracy and precision) and probabilistic (i.e., calibration and information) performance scores for the selected individual models

Model	Accuracy	Precision	Calibration	Information
A	1.18	2.12	0.0030	1.62
B	1.59	2.53	0.0094	2.13
C	0.95	2.17	0.0920	1.81
D	1.33	1.93	0.0010	2.45
E	1.34	1.91	0.0004	2.58
F	1.28	1.96	0.0013	2.45
G	1.30	1.93	0.0008	2.46
H	1.01	2.46	0.1191	1.13
I	1.09	2.08	0.2146	1.35
J	0.92	1.93	0.0272	1.43
K	0.95	1.74	<< 0.0001	1.93
L	0.92	1.79	0.0001	1.90
M	1.31	2.24	<< 0.0001	1.78
N	1.05	1.87	0.0777	1.50
O	0.82	2.08	0	5.97
P	1.20	1.88	0.0011	1.90
Q	2.01	3.55	0	2.93
R	1.30	1.89	0.0010	2.05
S	1.03	1.85	0.0010	1.84
T	1.31	2.13	0.0001	2.14
U	1.09	1.74	0.1366	1.34
V	1.02	1.74	<< 0.0001	2.32

Table S3. Median predictive (i.e., accuracy and precision) and probabilistic (i.e., calibration and information) performance scores for the ensemble models

Ensemble	Accuracy	Precision	Calibration	Information
CDC	1.11	1.67	0.20	1.58
CM	1.03	2.02	0.78	0.27
EWD	1.06	1.59	0.02	0.40

Abbreviations: CDC = Centers for Disease Control and Prevention; CM = Classical Model; EWD = Equal weights applied to densities

Tables S4. Probabilistic performance scores for the ensemble models by quantile (i.e., data for **Figure 4**)

Performance Measure	Percentile	Ensemble Weights		
		CDC	**EWD**	**CM**
Calibration	5%	0.0099	< 0.0001	0.2779
	25%	0.0543	0.0022	0.5610
	50%	0.2012	0.0165	0.7751
	75%	0.4442	0.0695	0.8851
	95%	0.7906	0.1995	0.9769
Information	5%	1.28	0.29	0.14
	25%	1.45	0.34	0.18
	50%	1.58	0.40	0.27
	75%	1.91	0.61	0.50
	95%	2.42	0.95	1.02

Abbreviations: CDC = Centers for Disease Control and Prevention; CM = Classical Model; EWD = Equal weights applied to densities

Table S5. Calibration scores of the ensemble models by state

State	Ensemble Weights		
	CDC	CM	EWD
Alabama	0.58	0.98	0.02
Alaska	0.03	0.80	0.21
Arizona	0.52	0.34	0.03
Arkansas	0.01	0.80	0.05
California	0.14	0.72	0.10
Colorado	0.22	0.94	0.05
Connecticut	0.97	0.76	0.02
Delaware	0.81	0.75	0.19
Florida	0.01	0.74	0.02
Georgia	0.10	0.80	0.01
Idaho	0.66	0.69	<< 0.01
Illinois	0.37	0.52	<< 0.01
Indiana	0.67	0.89	<< 0.01
Iowa	0.20	0.74	<< 0.01
Kansas	0.26	0.96	0.04
Kentucky	0.01	0.83	0.54
Louisiana	0.22	0.86	0.02
Maine	0.11	0.93	0.14
Maryland	0.33	0.38	<< 0.01
Massachusetts	0.38	0.13	0.07
Michigan	0.47	0.41	0.01
Minnesota	0.33	0.85	< 0.01
Mississippi	0.40	0.85	0.02
Missouri	0.76	0.84	0.08
Montana	0.50	0.26	0.10
Nebraska	0.02	0.93	0.39
Nevada	0.63	0.80	0.02
New Hampshire	0.04	0.32	< 0.01
New Jersey	0.18	0.78	<< 0.01
New Mexico	0.04	0.68	< 0.01
New York	0.43	0.98	0.06
North Carolina	0.25	0.61	< 0.01
North Dakota	0.02	0.68	< 0.01
Ohio	<< 0.01	0.10	0.01
Oklahoma	0.07	0.58	0.01
Oregon	0.42	0.98	0.16
Pennsylvania	0.02	0.56	0.01
Rhode Island	0.58	0.30	0.01
South Carolina	0.05	0.37	0.09
South Dakota	0.11	0.89	<< 0.01
Tennessee	0.20	0.96	0.01
Texas	0.02	0.79	0.14
Utah	0.11	0.68	0.01
Vermont	0.19	0.86	0.01
Virginia	0.06	0.35	<< 0.01
Washington	0.44	0.97	0.07
West Virginia	0.86	0.91	0.03
Wisconsin	0.03	0.94	<< 0.01
Wyoming	0.10	0.48	0.04
Calibration > a = 0.05	75.5%	100.0%	28.6%

Abbreviations: CDC = Centers for Disease Control and Prevention; CM = Classical Model; EWD = Equal weights applied to densities

Supplemental Figures

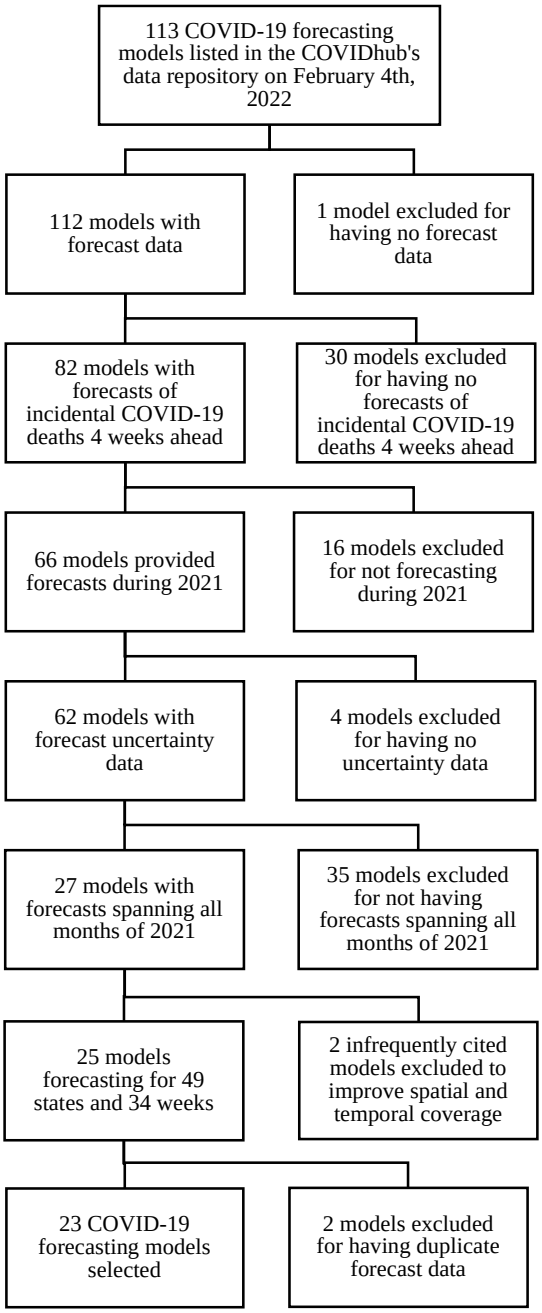


Figure S1. A diagram of our model selection process, with inclusion and exclusion criteria specified.